

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A Scenario-Based Research Assessment

Code of Conduct:

I RASHMITHA SENTHIL KUMAR (192525034) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	

Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

**ORBITAI: An AI-Powered Autonomous Space Debris
Hunter Satellite for Intelligent Detection, Tracking, and
Removal of Orbital Debris**

CO1- Assessment Tool 2

**Submitted in the partial fulfilment for the Course of
MLA0104-Artificial Intelligence and Expert System**

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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ABSTRACT

The rapid growth of satellite technology has significantly improved communication, navigation, weather forecasting, scientific research, and global connectivity. However, the increasing number of satellites has also resulted in the accumulation of space debris, including inactive satellites, rocket stages, and collision fragments. These objects travel at extremely high speeds and pose a serious threat to operational satellites and future space missions. Even a small piece of debris can cause severe damage due to its high orbital velocity, making space debris one of the major challenges in modern aerospace engineering. Current space surveillance systems mainly rely on ground-based radar, optical telescopes, and human analysis to monitor orbital objects and predict collision risks. Although these methods are effective for tracking debris, they require significant human intervention and do not provide an autonomous solution for debris removal. As the number of orbital objects continues to increase, there is a growing need for intelligent systems capable of automatically detecting, tracking, predicting, and removing hazardous debris.

This report proposes ORBITAI (Orbital Artificial Intelligence), an AI-powered autonomous satellite designed to improve space debris management. The proposed system integrates Artificial Intelligence, Computer Vision, Machine Learning, sensor fusion, and robotic technology to identify hazardous debris, predict collision risks, and determine the safest interception strategy. Using onboard cameras, radar, and LiDAR sensors, ORBITAI continuously monitors nearby objects, while AI algorithms analyze orbital data in real time to support autonomous decision-making. A robotic capture mechanism then safely removes dangerous debris by transferring it to a disposal orbit or guiding it toward controlled atmospheric re-entry. The proposed system aims to reduce collision risks, improve satellite safety, minimize human intervention, and support sustainable space operations. Although ORBITAI is presented as a conceptual solution, it is based on emerging technologies currently being explored in the aerospace industry. This project demonstrates how Artificial Intelligence can enhance autonomous space missions and contribute to the long-term protection of Earth's orbital environment.

Keywords: Artificial Intelligence (AI), Machine Learning (ML), Computer Vision, Reinforcement Learning, Space Debris, Autonomous Satellite, Sensor Fusion, Orbital Sustainability, Space Robotics.

1. INTRODUCTION

Space technology has become an essential part of modern society, supporting communication, navigation, weather forecasting, Earth observation, disaster management, scientific research, and national security. Thousands of satellites orbit the Earth every day, providing services that millions of people depend on. With the rapid growth of commercial and government space missions, the number of satellites launched into orbit has increased significantly over the past two decades.

Along with operational satellites, a large amount of space debris has also accumulated in Earth's orbit. Space debris refers to non-functional satellites, abandoned rocket stages, mission-related objects, and fragments generated from collisions or explosions. These objects continue to travel at extremely high speeds, often reaching 28,000 km/h. Due to their high velocity, even small debris particles can severely damage operational satellites or spacecraft, resulting in service interruptions, financial losses, and the creation of additional debris.

At present, space agencies use radar systems, optical telescopes, and satellite tracking networks to monitor orbital debris. These systems help identify potential collision risks and support mission planning. However, most existing approaches depend heavily on continuous human monitoring and manual decision-making. As the number of satellites and debris objects continues to grow, traditional monitoring methods face challenges in processing large volumes of data and responding quickly to potential threats.

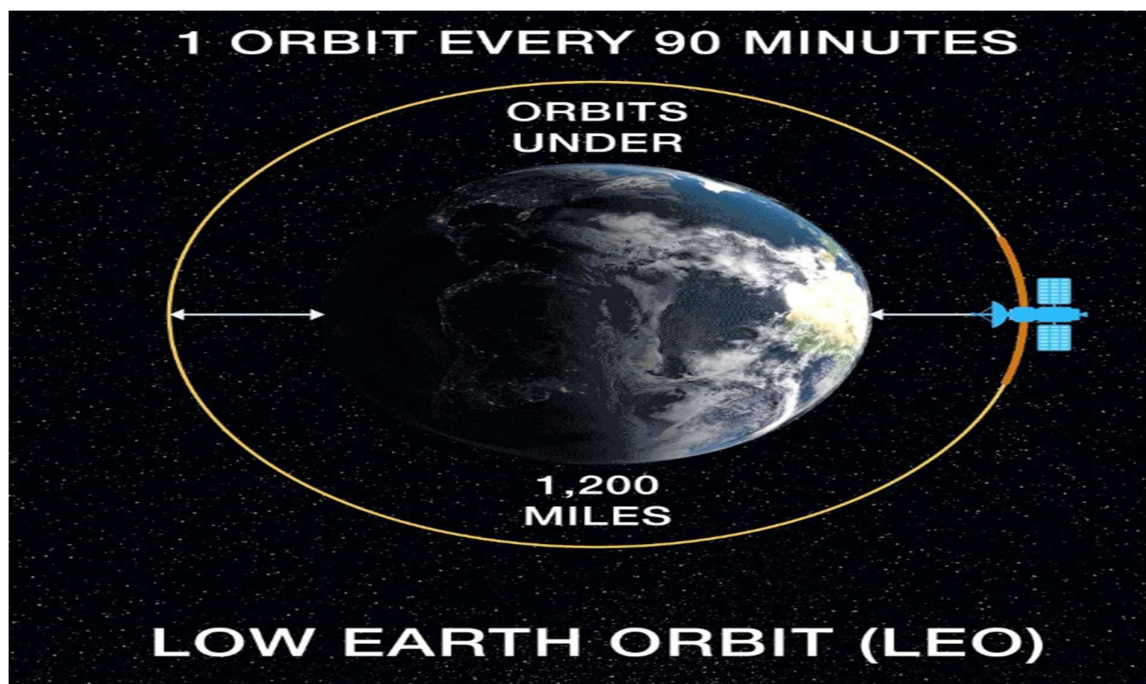
Artificial Intelligence (AI) offers an effective solution to these challenges by enabling autonomous monitoring, intelligent decision-making, and predictive analysis. AI algorithms can analyze sensor data in real time, identify hazardous debris, predict future orbital paths, estimate collision probabilities, and recommend or perform appropriate corrective actions. By integrating AI with advanced sensing technologies and robotic systems, future satellites can become capable of independently detecting and removing dangerous debris from space.

This report presents ORBITAI (Orbital Artificial Intelligence), a proposed AI-powered autonomous satellite designed for the detection, tracking, prediction, and removal of space debris. The proposed system combines Computer Vision, Machine Learning, Reinforcement Learning, sensor fusion, orbital trajectory prediction, and robotic capture mechanisms to create an intelligent debris management system. ORBITAI continuously monitors nearby orbital objects, analyzes their movement using AI models, identifies high-risk debris, and

autonomously plans safe interception and removal operations with minimal human intervention.

The primary objective of this project is to demonstrate how Artificial Intelligence can improve the safety, efficiency, and sustainability of space operations. By reducing collision risks and protecting valuable satellite infrastructure, ORBITAI contributes to the long-term preservation of Earth's orbital environment. This report discusses the proposed system architecture, AI technologies, workflow, advantages, challenges, and future scope, highlighting the growing role of AI in next-generation space exploration and autonomous satellite systems.

Figure 1. Illustration of Earth surrounded by satellites and orbital debris.



2. LITERATURE REVIEW

The growing challenge of space debris has attracted significant attention from researchers, space agencies, and aerospace industries worldwide. Several studies have focused on improving space situational awareness, orbital debris tracking, collision prediction, and autonomous debris removal using Artificial Intelligence (AI), robotics, and advanced sensing technologies. These research efforts provide the foundation for developing intelligent systems capable of ensuring the long-term sustainability of Earth's orbital environment.

The National Aeronautics and Space Administration (NASA) established the Orbital Debris Program Office to monitor and analyze objects orbiting Earth. NASA's research primarily focuses on identifying debris, predicting collision risks, and developing mitigation strategies to reduce the creation of additional orbital debris. Advanced orbital modeling and tracking algorithms are continuously used to improve collision prediction accuracy. However, NASA's current systems primarily support monitoring and avoidance rather than fully autonomous debris removal.

Similarly, the European Space Agency (ESA) has introduced the ClearSpace-1 mission, which represents one of the world's first active space debris removal missions. The project aims to capture a non-functional payload adapter using a robotic spacecraft equipped with autonomous guidance and robotic capture mechanisms. This mission demonstrates the feasibility of active debris removal; however, its decision-making capabilities are still largely mission-specific and supervised by ground control. The ORBITAI system extends this concept by integrating onboard Artificial Intelligence capable of autonomous object identification, risk assessment, mission planning, and robotic interception with minimal human intervention.

Another important contribution comes from Astroscale Holdings Inc., a private aerospace company developing the End-of-Life Services by Astroscale (ELSA-d) mission. The ELSA-d demonstration successfully validated autonomous rendezvous, proximity operations, magnetic docking, and satellite servicing technologies. Although Astroscale focuses primarily on servicing and deorbiting client satellites, its research highlights the increasing role of autonomous robotic systems in future orbital maintenance missions. ORBITAI incorporates similar autonomous navigation concepts while enhancing them through AI-driven decision-making and predictive analytics.

Recent advances in Artificial Intelligence have significantly improved the ability of satellites to process sensor data onboard. Computer Vision algorithms such as Convolutional Neural Networks (CNNs) and YOLO (You Only Look Once) enable rapid detection and classification of orbital objects using optical imagery. These models can accurately distinguish active satellites from inactive debris, reducing false detections and improving overall monitoring efficiency. Researchers have demonstrated that deep learning models outperform conventional image-processing techniques in detecting small and irregularly shaped orbital objects.

Machine Learning techniques have also been applied to orbital trajectory prediction and collision probability estimation. Algorithms such as the Kalman Filter, Long Short-Term Memory (LSTM) networks, and Bayesian estimation methods analyze historical orbital data and continuously update object trajectories based on real-time observations. These predictive models improve collision forecasting accuracy and enable earlier intervention compared with traditional orbital mechanics alone.

In addition, Reinforcement Learning (RL) has emerged as a promising approach for autonomous spacecraft navigation and mission planning. RL-based agents learn optimal maneuvering strategies through continuous interaction with simulated orbital environments, minimizing fuel consumption while maximizing mission success. Such intelligent decision-making techniques are particularly valuable for autonomous debris interception missions where dynamic orbital conditions require adaptive responses.

Another key research area is Sensor Fusion, which combines information from multiple sensors such as optical cameras, radar, LiDAR, Global Positioning System (GPS), and Inertial Measurement Units (IMU). Sensor fusion significantly improves detection reliability by compensating for the limitations of individual sensors, reducing uncertainty, and providing a comprehensive understanding of the orbital environment. This approach has become a fundamental component of autonomous aerospace systems.

Despite these advancements, current research still faces several challenges. Most existing systems focus either on debris tracking or mission-specific debris removal rather than providing an integrated AI-driven autonomous solution capable of continuous monitoring, intelligent decision-making, robotic capture, and safe debris disposal. Furthermore, many existing approaches rely heavily on continuous communication with ground stations, limiting their ability to respond rapidly during time-critical situations.

The proposed ORBITAI system addresses these research gaps by integrating Computer Vision, Machine Learning, Reinforcement Learning, Sensor Fusion, Edge AI, Autonomous Navigation, and Robotic Capture Technologies into a unified intelligent framework. Unlike conventional systems, ORBITAI is designed to autonomously detect hazardous debris, predict collision risks, determine optimal interception paths, capture dangerous orbital objects, and safely dispose of them with minimal human intervention. By combining recent advances in Artificial Intelligence with modern aerospace technologies, ORBITAI represents a future-ready solution for sustainable space debris management and next-generation autonomous satellite operations.

Table 2.1 Literature Comparison

Research / Organization	Main Contribution	Limitation	ORBITAI Improvement
NASA Orbital Debris Program	Space debris tracking and collision prediction	No autonomous debris removal	AI-driven autonomous removal
ESA ClearSpace-1	Active robotic debris capture	Mission-specific operation	Intelligent onboard AI decision-making
Astroscale ELSA-d	Autonomous satellite servicing	Focused on client satellites	Universal debris identification and removal
AI Computer Vision Research	Automatic object detection	Image-based only	Multi-sensor AI detection
Reinforcement Learning Research	Autonomous navigation	Simulation-focused	Real-time orbital mission planning

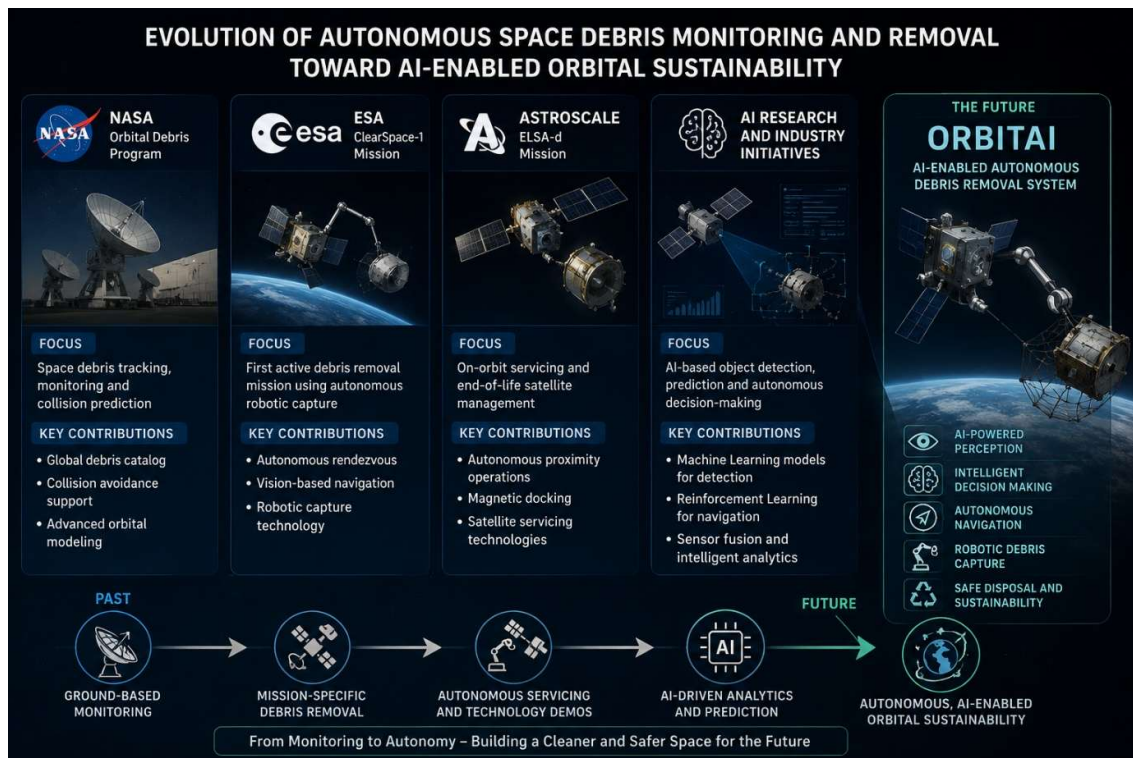


Figure 2.1: Current research and industrial initiatives in autonomous space debris monitoring and removal, highlighting the evolution toward AI-enabled orbital sustainability.

3.PROBLEM STATEMENT

The rapid increase in satellite launches has significantly contributed to the accumulation of space debris in Earth's orbit. Thousands of inactive satellites, abandoned rocket stages, and collision fragments continue to travel at extremely high speeds, posing a serious threat to operational satellites, space missions, and future space exploration. Even small debris particles can cause catastrophic damage due to their high orbital velocity, leading to satellite failures, communication disruptions, and the creation of additional debris through collisions. Existing space debris monitoring systems primarily rely on ground-based radar, optical telescopes, and manual analysis to detect and track orbital objects. Although these systems provide valuable information, they require continuous human intervention for collision assessment, decision-making, and mission planning. As the number of space objects continues to grow, traditional monitoring methods face challenges in processing large volumes of data, predicting collision risks efficiently, and responding quickly to potential threats.

There is a need for an intelligent and autonomous system capable of continuously monitoring orbital debris, accurately predicting collision risks, prioritizing hazardous objects, and performing debris removal with minimal human intervention. Integrating Artificial Intelligence with advanced sensing technologies and robotic systems can improve the speed, accuracy, and efficiency of space debris management while supporting the long-term sustainability of Earth's orbital environment. The proposed ORBITAI system addresses these challenges by utilizing Artificial Intelligence, Computer Vision, Machine Learning, and autonomous robotic technology to detect, track, predict, and safely remove hazardous space debris, thereby enhancing satellite safety and ensuring the sustainability of future space operations.

4. OBJECTIVES

The primary objective of the proposed ORBITAI (Orbital Artificial Intelligence) system is to develop an intelligent and autonomous satellite capable of detecting, tracking, predicting, and removing hazardous space debris from Earth's orbit using advanced Artificial Intelligence techniques. The proposed system aims to enhance the safety and sustainability of space operations by continuously monitoring the orbital environment through high-resolution cameras, radar, LiDAR, and other onboard sensors. Another important objective is to employ Computer Vision and Machine Learning algorithms to accurately identify and classify orbital objects, distinguishing active satellites from inactive debris with high precision. The system is also designed to predict the future trajectory of detected objects and estimate collision probabilities using AI-based predictive models, enabling early identification of potential threats. Furthermore, ORBITAI seeks to minimize human intervention by incorporating autonomous decision-making capabilities, allowing the satellite to independently prioritize hazardous debris, determine the safest interception strategy, and execute debris removal operations efficiently. The proposed system also aims to optimize satellite navigation by selecting fuel-efficient orbital paths and utilizing a robotic capture mechanism to safely collect and remove dangerous debris. In addition, ORBITAI is intended to reduce the operational risks and economic losses associated with satellite collisions while improving the overall reliability and lifespan of space infrastructure. Another objective is to demonstrate the practical integration of Artificial Intelligence, Machine Learning, Computer Vision, Reinforcement Learning, Sensor Fusion, Autonomous Robotics, and Aerospace Engineering into a single intelligent system capable of solving a complex real-world problem. Finally, this project aims to contribute toward the long-term sustainability of space exploration by reducing orbital congestion, preventing the generation of additional debris, and promoting the development of next-generation autonomous satellite technologies that support safe, efficient, and responsible utilization of Earth's orbital environment.

5.EXISTING SYSTEM

The increasing amount of space debris has become a major concern for space agencies worldwide. To minimize the risk of collisions, organizations such as NASA, the European Space Agency (ESA), and other national space agencies operate Space Surveillance and Tracking (SST) systems to monitor objects orbiting the Earth. These systems primarily use ground-based radar stations, optical telescopes, and space-based sensors to detect, track, and maintain a catalog of active satellites and orbital debris.

When a potential collision is detected, the collected data is analyzed by mission control centers using orbital prediction software. Engineers calculate the probability of collision and determine whether a satellite should perform a collision avoidance maneuver. If the collision risk exceeds a predefined threshold, commands are transmitted from ground stations to the satellite, allowing it to change its orbit and avoid impact. Although this approach has successfully prevented many satellite collisions, it relies heavily on human expertise and continuous monitoring.

The existing system mainly focuses on debris detection and collision prediction rather than active debris removal. Once hazardous debris is identified, it continues to remain in orbit, posing future risks to other spacecraft. As the number of satellites and orbital debris continues to increase, manual monitoring becomes more complex, time-consuming, and computationally demanding. Furthermore, communication delays between ground stations and satellites reduce the ability to respond instantly to rapidly changing situations.

Therefore, while current space surveillance systems provide valuable monitoring capabilities, they have limited autonomous decision-making and do not offer an intelligent solution for detecting, prioritizing, capturing, and removing hazardous debris from Earth's orbit. These limitations highlight the need for an AI-powered autonomous system such as ORBITAI, which can continuously monitor orbital objects, make intelligent decisions in real time, and actively remove dangerous space debris with minimal human intervention.

6.LIMITATIONS OF EXISTING SYSTEM

Although current space surveillance and tracking systems have significantly improved the monitoring of orbital objects, they still face several limitations in managing the growing amount of space debris. Most existing systems are designed to detect and track debris rather than actively remove it. As the number of satellites and debris fragments increases, these limitations become more critical and may affect the safety and sustainability of future space missions.

One of the major limitations is the heavy dependence on human intervention. Collision risk analysis, mission planning, and avoidance decisions are primarily performed by engineers at ground control stations. This process requires continuous monitoring and can delay responses during time-critical situations.

Another significant limitation is the absence of autonomous debris removal. Existing systems can predict potential collisions and recommend orbit adjustments for operational satellites, but they cannot capture or eliminate hazardous debris. As a result, dangerous objects remain in orbit and continue to threaten future missions.

Current monitoring systems also generate large volumes of orbital data, making real-time analysis increasingly difficult. As thousands of new satellites are launched every year, traditional processing methods struggle to efficiently analyze, prioritize, and respond to potential threats.

In addition, communication delays between satellites and ground stations can reduce the speed of decision-making. During unexpected orbital changes, relying solely on ground-based control may not provide the rapid response required to prevent collisions.

Furthermore, existing systems have limited capability to detect very small debris particles, which can still cause severe damage because of their extremely high orbital velocity. Weather conditions, atmospheric disturbances, and the geographical coverage of ground-based observation systems may also reduce tracking accuracy.

These limitations highlight the need for an intelligent and autonomous solution capable of continuously monitoring orbital objects, making real-time decisions, predicting collision risks, and actively removing hazardous debris without depending entirely on human operators.

7.PROPOSED SYSTEM – ORBITAI

To overcome the limitations of existing space debris monitoring systems, this report proposes ORBITAI (Orbital Artificial Intelligence), an AI-powered autonomous satellite designed to detect, track, predict, and remove hazardous space debris with minimal human intervention. Unlike conventional systems that mainly monitor debris from ground stations, ORBITAI performs intelligent decision-making directly in space using onboard Artificial Intelligence.

The proposed system integrates multiple advanced technologies, including Computer Vision, Machine Learning, Reinforcement Learning, Sensor Fusion, Orbital Trajectory Prediction, and Autonomous Robotics, to create a complete debris management solution. High-resolution cameras, LiDAR, radar, and other onboard sensors continuously scan the surrounding orbital environment and collect real-time data. The collected information is processed by AI algorithms that identify debris, classify objects, estimate their size and velocity, and predict future orbital trajectories. Based on the predicted collision probability, the AI Decision Engine prioritizes the most dangerous debris and determines the safest interception strategy. ORBITAI then autonomously navigates toward the target object using an optimized orbital path. A robotic capture mechanism, such as a robotic arm or deployable smart capture net, securely captures the debris. Finally, the captured object is either transferred to a designated disposal orbit or guided toward Earth's atmosphere for controlled atmospheric re-entry, where it safely burns up. The proposed system significantly reduces dependence on ground control by enabling autonomous monitoring, intelligent decision-making, and real-time response. ORBITAI not only improves satellite safety but also contributes to the long-term sustainability of Earth's orbital environment by actively removing hazardous debris before collisions occur.

Unlike existing space debris monitoring systems, ORBITAI integrates Computer Vision, Reinforcement Learning, Sensor Fusion, Edge AI, Autonomous Navigation, and Robotic Capture into a unified intelligent framework capable of end-to-end autonomous debris detection, decision-making, interception, and safe disposal.

8.USE CASE

One practical use case of ORBITAI involves protecting an operational communication satellite from a potential collision with a large fragment of space debris.

As ORBITAI continuously monitors nearby orbital objects, its onboard sensors detect an unidentified object moving toward an active communication satellite. The AI system immediately analyzes the collected sensor data, identifies the object as inactive space debris, and predicts its orbital trajectory. After calculating the collision probability, the system determines that the debris poses a high risk to the operational satellite.

The AI Decision Engine automatically prioritizes the threat and generates the safest interception route. ORBITAI adjusts its orbit, approaches the debris, and activates its robotic capture mechanism to safely secure the object. Once captured, the debris is transported to a disposal orbit or directed toward controlled atmospheric re-entry, eliminating the collision risk.

Throughout the mission, ORBITAI continuously updates mission data and communicates important information to the ground control station. Since most decisions are performed autonomously onboard, the response time is significantly reduced compared to conventional monitoring systems.

This use case demonstrates how Artificial Intelligence can transform space debris management from passive monitoring into an intelligent, autonomous, and proactive protection system.

9.WHY THIS USE CASE?

The selected use case addresses one of the most critical challenges in modern space exploration—the increasing risk of collisions caused by space debris. Every year, thousands of new satellites are launched into Earth's orbit, while existing debris continues to accumulate. Traditional systems can predict potential collisions but cannot actively remove hazardous objects, allowing the risk to persist.

The ORBITAI use case demonstrates how Artificial Intelligence can move beyond simple monitoring by enabling autonomous detection, intelligent risk assessment, and active debris removal. This approach reduces dependence on human operators, improves response speed, and minimizes the possibility of catastrophic satellite collisions.

The proposed system also supports the sustainable use of space by preventing the formation of additional debris caused by collisions. Protecting communication satellites, weather satellites, navigation systems, and scientific spacecraft helps ensure the continuous availability of services that modern society depends upon.

Furthermore, this use case highlights the practical integration of Artificial Intelligence, Machine Learning, Robotics, Computer Vision, and Aerospace Engineering into a single autonomous system. It demonstrates how AI can solve complex real-world problems that require rapid decision-making, predictive analysis, and autonomous action.

Therefore, ORBITAI is not only an innovative research concept but also a future-ready solution that has the potential to improve the safety, reliability, and sustainability of space operations.

10. TECHNICAL ARCHITECTURE

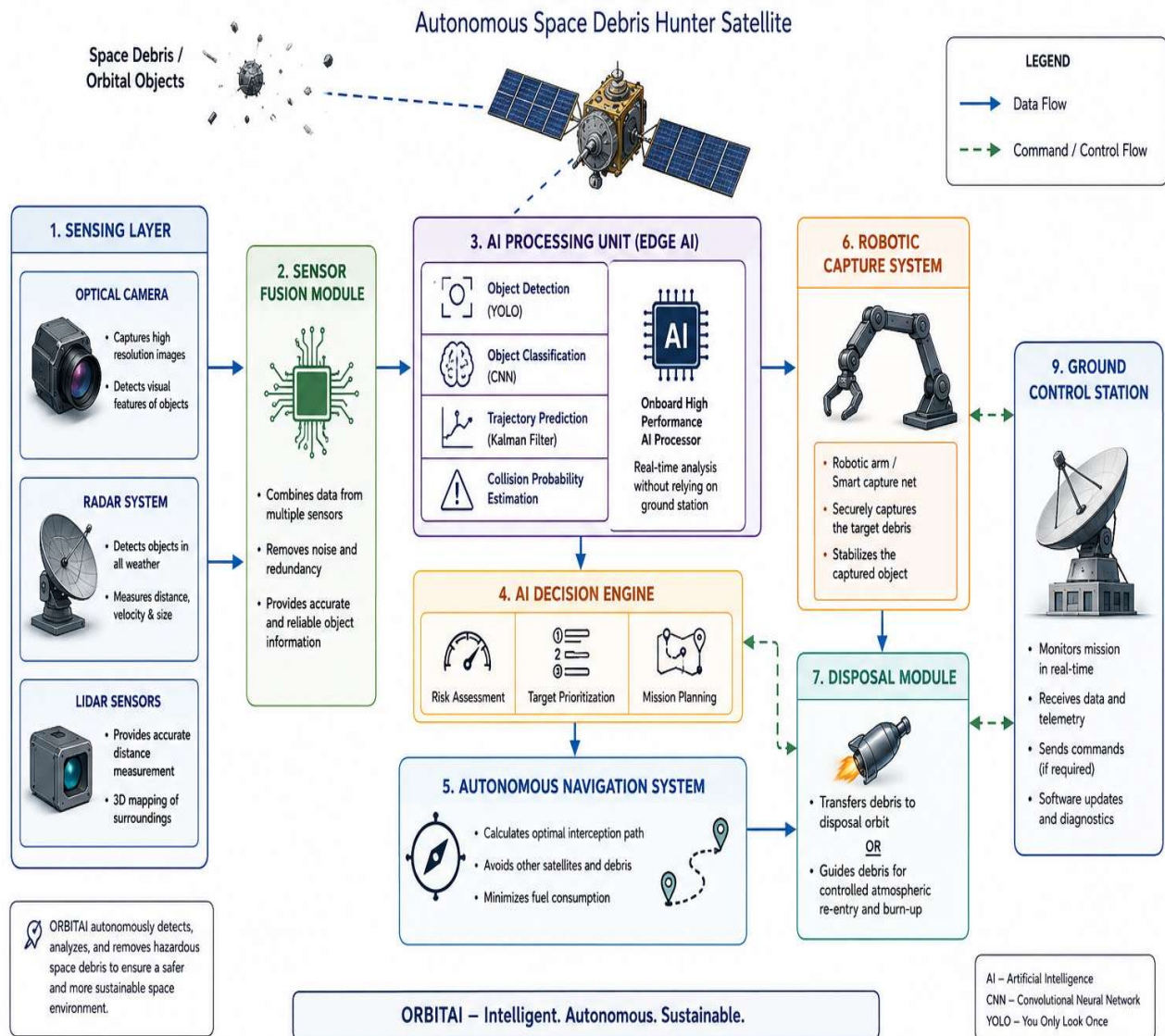


Figure 10.1: Technical architecture of the proposed ORBITAI system illustrating sensing, AI processing, autonomous navigation, robotic capture, disposal, and ground control modules.

11. WORKING MECHANISM OF ORBITAI

The ORBITAI system operates through a sequence of intelligent and autonomous processes that enable the satellite to detect, analyze, intercept, and remove hazardous space debris with minimal human intervention. The working mechanism integrates Artificial Intelligence, Machine Learning, Computer Vision, autonomous navigation, and robotic technology to perform each stage of the mission efficiently. The complete operational process is illustrated below.

Step 1: Space Environment Monitoring

ORBITAI continuously observes the surrounding orbital environment using high-resolution optical cameras, radar, LiDAR sensors, GPS, and Inertial Measurement Units (IMU). These sensors collect real-time information about nearby satellites, orbital debris, relative velocity, distance, and object position. Continuous monitoring ensures that newly detected objects are immediately analyzed without waiting for instructions from ground control.

Step 2: Data Acquisition and Sensor Fusion

The collected sensor data is transmitted to the onboard Sensor Fusion Module. Since each sensor has different strengths and limitations, Sensor Fusion combines information from all available sensors into a single accurate dataset. This process eliminates redundant information, reduces measurement noise, improves object localization, and increases overall detection reliability.

Step 3: AI-Based Object Detection

The fused data is processed by the Edge AI Computing Unit using Computer Vision and Deep Learning algorithms. Artificial Intelligence automatically detects every object within the satellite's observation range and extracts important features such as object size, shape, motion, brightness, and orbital characteristics.

Step 4: Object Classification

After detection, Machine Learning models classify each object into one of the following categories:

- Active Satellite
- Inactive Satellite

- Rocket Body
- Collision Fragment
- Hazardous Space Debris

This classification enables the system to distinguish valuable operational satellites from dangerous orbital debris.

Step 5: Trajectory Prediction and Collision Analysis

The AI Prediction Engine continuously estimates the future orbital trajectory of every detected object. By analyzing orbital velocity, altitude, direction, and historical motion patterns, the system predicts possible collision scenarios. If the calculated collision probability exceeds the predefined safety threshold, the object is marked as a high-priority target.

Step 6: Intelligent Decision Making

The AI Decision Engine evaluates multiple parameters, including collision probability, debris size, mission priority, available fuel, orbital distance, and interception feasibility. Based on these factors, the system autonomously determines whether debris removal is necessary and selects the safest interception strategy.

Step 7: Autonomous Navigation

Once the mission is approved, the Autonomous Navigation Module calculates the most fuel-efficient interception path. Artificial Intelligence continuously updates the navigation plan based on changing orbital conditions while avoiding interference with nearby operational satellites.

Step 8: Robotic Debris Capture

After reaching the target, the Robotic Capture System deploys a robotic arm or an intelligent capture net to securely capture the debris. The robotic mechanism stabilizes the object before initiating the disposal procedure.

Step 9: Safe Disposal

The captured debris is transported to a designated disposal orbit or guided toward Earth's atmosphere, where it safely burns during controlled atmospheric re-entry. This process permanently eliminates the collision risk without generating additional debris.

Step 10: Mission Reporting and Continuous Learning

Upon successful mission completion, ORBITAI automatically generates mission reports containing debris characteristics, AI decisions, fuel consumption, capture performance, and mission duration. These reports are transmitted to the Ground Control Station for monitoring. The onboard AI continuously updates its learning models using mission data, improving future detection accuracy and autonomous decision-making capabilities.

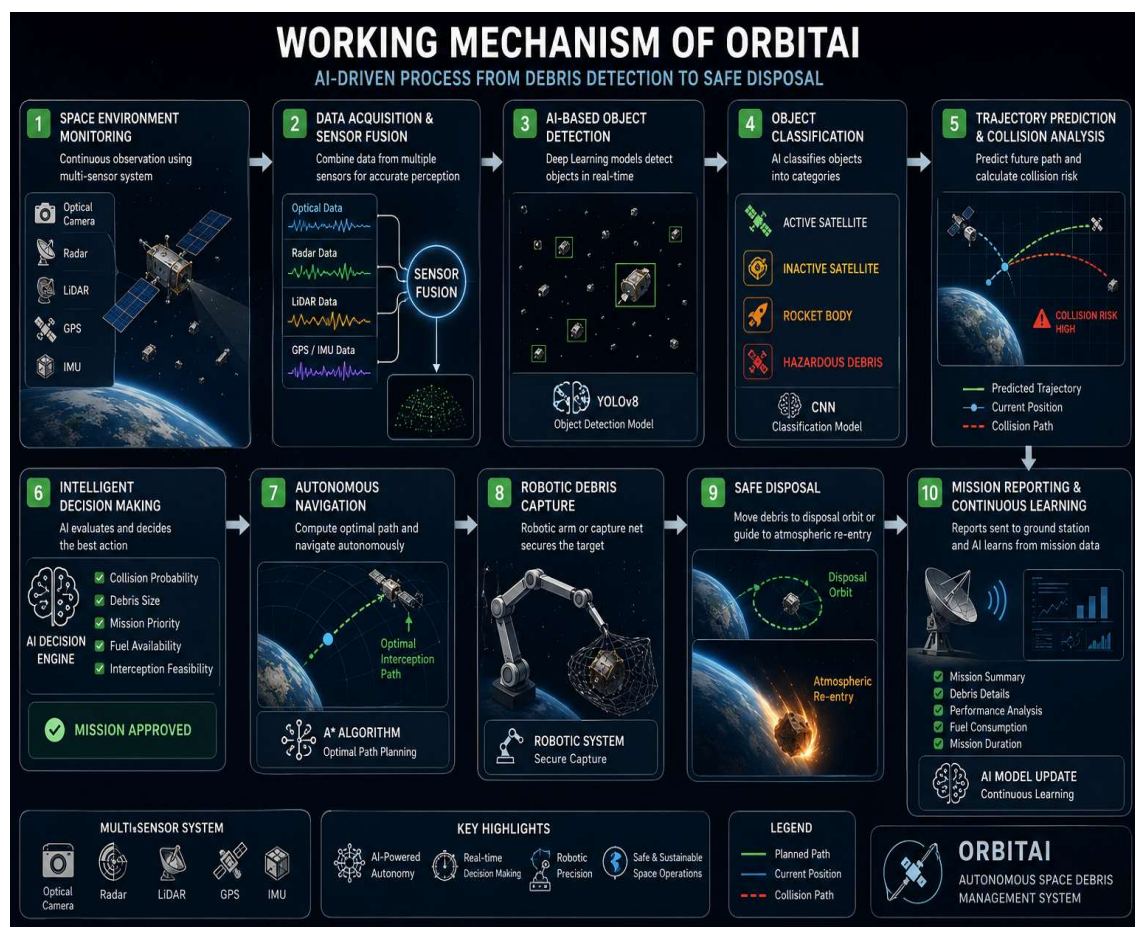


Figure 11.1: Working mechanism of the proposed ORBITAI system illustrating the complete AI-driven process from debris detection to safe disposal.

Table 11.1: Working Mechanism Summary

Stage	AI Module	Output
Space Monitoring	Sensors	Real-time orbital data
Data Processing	Sensor Fusion	Integrated sensor information
Object Detection	YOLO	Debris detection
Classification	CNN	Satellite or debris classification
Prediction	Kalman Filter	Future trajectory estimation
Decision	Deep Q Network	Mission approval
Navigation	A* Algorithm	Optimal interception path
Capture	Robotic System	Debris secured
Disposal	Autonomous Control	Safe debris removal
Reporting	Ground Station	Mission analysis

12. DATA SOURCES AND DATA PROCESSING

Artificial Intelligence systems require accurate and reliable data to make intelligent decisions. The proposed ORBITAI system utilizes data from multiple onboard sensors and external space surveillance databases to monitor the orbital environment continuously. By integrating real-time sensor observations with historical orbital information, the system can accurately detect hazardous debris, predict collision risks, and autonomously plan debris removal missions. The combination of diverse data sources and intelligent data processing significantly improves the reliability and efficiency of the proposed system.

12.1 Data Sources

The ORBITAI system collects information from both onboard sensing devices and external orbital databases. These data sources provide comprehensive information about satellites, orbital debris, object trajectories, and environmental conditions.

1. Optical Camera

High-resolution optical cameras continuously capture images of nearby orbital objects. These images are used by Computer Vision algorithms to detect, recognize, and classify satellites and space debris.

2. Radar Sensor

Radar sensors detect objects regardless of lighting conditions or orbital shadows. Radar provides accurate measurements of object distance, relative velocity, and movement direction.

3. LiDAR Sensor

LiDAR (Light Detection and Ranging) generates precise three-dimensional measurements of nearby objects. It estimates object size, shape, surface characteristics, and relative position with high accuracy.

4. GPS and Inertial Measurement Unit (IMU)

GPS and IMU provide the satellite's current position, orientation, speed, and attitude information. These measurements support autonomous navigation and precise orbital maneuvering.

5. NASA Orbital Debris Database

Historical orbital information, debris catalogs, and collision records published by NASA are used for AI model training and orbital prediction.

6. ESA Space Surveillance Database

The European Space Agency provides information about tracked orbital objects, collision warnings, and orbital characteristics that improve AI prediction accuracy.

7. Historical Mission Data

Data collected from previous ORBITAI missions are stored within the onboard database. Machine Learning algorithms continuously learn from historical observations to improve future decision-making.

12.2 Data Processing

After collecting information from multiple sources, ORBITAI performs several processing stages before making autonomous decisions.

Step 1: Data Collection

Sensor data is continuously collected from optical cameras, radar, LiDAR, GPS, and IMU. Each sensor captures different characteristics of nearby orbital objects.

Step 2: Data Pre-processing

Raw sensor data often contains noise, redundant measurements, missing values, and environmental disturbances. During preprocessing, the system removes noise, filters unnecessary information, normalizes sensor values, and synchronizes data collected from different sensors.

Step 3: Sensor Fusion

The Sensor Fusion Module combines information from multiple sensors into a single high-confidence dataset. This process improves object localization, reduces uncertainty, and increases detection reliability.

Step 4: Feature Extraction

Artificial Intelligence extracts important object characteristics such as:

- Object Size
- Shape
- Relative Velocity
- Orbital Altitude
- Motion Direction
- Brightness
- Surface Reflection

These features become the input for Machine Learning models.

Step 5: AI Analysis

Deep Learning and Computer Vision algorithms analyze the extracted features to:

- Detect orbital objects
- Classify satellites and debris
- Estimate collision probability
- Predict future orbital trajectories

Step 6: Decision Generation

The AI Decision Engine evaluates all available information and determines whether debris removal is required. If necessary, it automatically generates the safest interception strategy.

Step 7: Mission Execution

The navigation module and robotic capture system execute the planned mission autonomously. All mission information is stored for continuous AI learning.

Table 12.1: Data Sources Used in ORBITAI

Data Source	Information Collected	Purpose
Optical Camera	Images of orbital objects	Object detection
Radar	Distance and velocity	Object tracking
LiDAR	3D object geometry	Size and position estimation
GPS & IMU	Satellite position and orientation	Navigation
NASA Database	Historical debris information	AI model training
ESA Database	Orbital tracking information	Collision prediction
Mission Database	Previous mission records	Continuous AI learning

13.ADVANTAGES

The proposed ORBITAI system offers several advantages over traditional space debris monitoring systems.

- Autonomous detection and tracking of orbital debris.
- Real-time collision prediction using Artificial Intelligence.
- Reduced dependence on continuous human monitoring.
- Faster decision-making through onboard Edge AI.
- Active removal of hazardous space debris instead of only monitoring it.
- Improved protection of communication, navigation, weather, and scientific satellites.
- Reduced probability of chain-reaction collisions (Kessler Syndrome).
- Optimized fuel consumption through intelligent path planning.
- Scalable architecture suitable for future satellite constellations.
- Supports sustainable and safer space exploration.

14. CHALLENGES

Although ORBITAI provides an advanced solution, several technical and operational challenges must be addressed.

- Designing AI systems that operate reliably in the harsh space environment.
- Limited onboard computing power and energy availability.
- Communication delays between satellites and ground stations.
- High development, launch, and maintenance costs.
- Precise robotic capture of fast-moving debris.
- Difficulty in detecting extremely small debris fragments.
- Need for highly accurate orbital trajectory prediction.
- Compliance with international space regulations and mission safety standards.
- Continuous software updates and cybersecurity protection for autonomous satellites.

Despite these challenges, ongoing advancements in Artificial Intelligence, robotics, satellite technology, and aerospace engineering are expected to make autonomous space debris removal systems increasingly practical in the future.

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15. FUTURE SCOPE

Future versions of ORBITAI can be enhanced by integrating advanced Artificial Intelligence and next-generation space technologies to improve the efficiency and autonomy of space debris management. AI swarm satellites can collaborate to detect, track, and remove multiple debris objects simultaneously, significantly reducing mission time and improving overall coverage. The incorporation of Digital Twin technology will enable real-time simulation and monitoring of satellite operations, allowing the system to optimize mission planning and predict potential failures before they occur. Advanced Quantum AI optimization techniques can further enhance trajectory planning, collision prediction, and fuel-efficient navigation for complex orbital environments.

In addition, future ORBITAI missions may support autonomous on-orbit servicing, satellite refueling, and maintenance operations, thereby extending the operational lifespan of valuable space assets. The integration of Space Traffic Management (STM) systems will facilitate safe coordination among thousands of satellites operating in Earth's orbit. Furthermore, Federated Learning can enable multiple satellites to collaboratively improve AI models by sharing knowledge instead of raw data, ensuring continuous learning while reducing communication overhead and enhancing data security.

With continuous advancements in Artificial Intelligence, robotics, edge computing, and aerospace engineering, ORBITAI has the potential to evolve into a fully autonomous orbital maintenance platform capable of protecting satellite infrastructure, minimizing orbital congestion, and supporting the long-term sustainability of space exploration. This future vision demonstrates the scalability and practical significance of the proposed system in addressing one of the most critical challenges in modern aerospace engineering.

16.CONCLUSION

The rapid growth of satellite launches has significantly increased the amount of space debris orbiting Earth, creating serious challenges for satellite safety and the long-term sustainability of space exploration. Existing space surveillance systems are effective in detecting and tracking orbital objects; however, they rely heavily on human intervention and do not provide an autonomous solution for debris removal. As the number of satellites and debris fragments continues to increase, there is a growing need for intelligent systems capable of making real-time decisions and actively mitigating collision risks.

This report proposed ORBITAI (Orbital Artificial Intelligence), an AI-powered autonomous Space Debris Hunter Satellite designed to detect, classify, predict, and remove hazardous space debris with minimal human intervention. The proposed system integrates Artificial Intelligence, Machine Learning, Computer Vision, Reinforcement Learning, Sensor Fusion, and Autonomous Robotics to create a comprehensive debris management solution. By combining real-time sensing with intelligent decision-making and robotic capture mechanisms, ORBITAI can efficiently identify high-risk debris, calculate safe interception paths, and remove dangerous objects from Earth's orbit.

The proposed architecture offers several advantages over conventional monitoring systems, including faster decision-making, improved collision prediction, reduced operational costs, enhanced satellite protection, and greater mission autonomy. Furthermore, ORBITAI contributes to the long-term sustainability of space operations by reducing orbital congestion and minimizing the risk of future collisions.

Although the proposed system is presented as a conceptual solution, it is inspired by current advancements in autonomous spacecraft, Artificial Intelligence, and space robotics. Future developments in AI algorithms, onboard computing, satellite communication, and robotic technologies will further improve the feasibility and performance of intelligent debris removal missions.

In conclusion, ORBITAI demonstrates the transformative potential of Artificial Intelligence in addressing one of the most critical challenges in modern aerospace engineering. The proposed system represents a future-ready approach for protecting valuable satellite infrastructure, ensuring safer space operations, and supporting sustainable exploration of Earth's orbital environment for future generations.

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Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (60–50%)
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.